Use Cases

for

Dialysis Transport Coordination

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Joshua Chin Wei Jie	18/09/2026	Added existing use cases	1.0
Joshua Chin Wei Jie	29/09/2026	Added new sub-use cases and updated existing use cases	1.1
Joshua Chin Wei Jie	17/10/2026	Updated existing use case	1.2
Joshua Chin Wei Jie	20/10/2026	Updated existing use cases	1.3

Guidance for Use Case Template

Document each use case using the template shown in the Appendix. This section provides a description of each section in the use case template.

1. Use Case Identification

1.1. Use Case ID

Give each use case a unique numeric identifier, in hierarchical form: X.Y. Related use cases can be grouped in the hierarchy. Functional requirements can be traced back to a labeled use case.

1.2. Use Case Name

State a concise, results-oriented name for the use case. These reflect the tasks the user needs to be able to accomplish using the system. Include an action verb and a noun. Some examples:

- View part number information.
- Manually mark hypertext source and establish link to target.
- Place an order for a CD with the updated software version.

1.3. Use Case History

1.3.1 Created By

Supply the name of the person who initially documented this use case.

1.3.2 Date Created

Enter the date on which the use case was initially documented.

1.3.3 Last Updated By

Supply the name of the person who performed the most recent update to the use case description.

1.3.4 Date Last Updated

Enter the date on which the use case was most recently updated.

2. Use Case Definition

2.1. Actor

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the actor(s) that will be performing this use case.

2.2. Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

2.3. Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. Number each precondition. Examples:

1. User's identity has been authenticated.
2. User's computer has sufficient free memory available to launch task.

2.4. Postconditions

Describe the state of the system at the conclusion of the use case execution. Number each postcondition. Examples:

1. Document contains only valid SGML tags.
2. Price of item in database has been updated with new value.

2.5. Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. The priority scheme used must be the same as that used in the software requirements specification.

2.6. Frequency of Use

Estimate the number of times this use case will be performed by the actors per some appropriate unit of time.

2.7. Flow of Events

Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. This description may be written as an answer to the hypothetical question, "How do I <accomplish the task stated in the use case name>?" This is best done as a numbered list of actions performed by the actor, alternating with responses provided by the system.

2.8. Alternative Flows

Document other, legitimate usage scenarios that can take place within this use case separately in this section. State the alternative course, and describe any differences in the sequence of steps that take place. Number each alternative course using the Use Case ID as a prefix, followed by "AC" to indicate "Alternative Course". Example: X.Y.AC.1.

2.9. Exceptions

Describe any anticipated error conditions that could occur during execution of the use case, and define how the system is to respond to those conditions. Also, describe how the system is to respond if the use case execution fails for some unanticipated reason. Number each exception using the Use Case ID as a prefix, followed by "EX" to indicate "Exception". Example: X.Y.EX.1.

2.10. Includes

List any other use cases that are included (“called”) by this use case. Common functionality that appears in multiple use cases can be split out into a separate use case that is included by the ones that need that common functionality.

2.11. Special Requirements

Identify any additional requirements, such as nonfunctional requirements, for the use case that may need to be addressed during design or implementation. These may include performance requirements or other quality attributes.

2.12. Assumptions

List any assumptions that were made in the analysis that led to accepting this use case into the product description and writing the use case description.

2.13. Notes and Issues

List any additional comments about this use case or any remaining open issues or TBDs (To Be Determined) that must be resolved. Identify who will resolve each issue, the due date, and what the resolution ultimately is.

Use Case Descriptions

UC-01: Authenticate User

Use Case ID	UC-01		
Use Case Name	Authenticate User		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Transport Coordinator, Driver, Patient, Caregiver		
Description	Allows a registered user to securely log in and access only the functions permitted by their assigned role.		
Preconditions	**1.** A registered user account exists. **2.** The login service is available.		
Postconditions	**1.** The user is authenticated. **2.** The user is granted access appropriate to their assigned role.		
Priority	High		
Frequency of Use	Once per user session.		
Flow of Events	**1.** User opens the login interface. **2.** User enters credentials. **3.** System validates the credentials. **4.** System identifies the user's assigned role. **5.** System grants authorised access.		
Alternative Flows	**UC-01.AF1 - Retry login:** If the user enters incorrect credentials, the system informs the user and allows them to enter their credentials again.		
Exceptions	**UC-01.EX1 - Account unavailable:** If the user's account is inactive or the user is unauthorised, access to the system is denied.		
Includes	None.		
Special Requirements	All protected functions require authenticated access and Role-Based Access Control (RBAC).		
Assumptions	User accounts and roles have already been created and assigned.		
Notes and Issues	Related to FR-1.2, FR-1.3, NFR-5.1 and NFR-5.2.		

UC-02: Manage Profiles & Resources

Use Case ID	UC-02		
Use Case Name	Manage Profiles & Resources		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	20/10/2026
Actor	Patient, Caregiver, Driver, Transport Coordinator		
Description	Allows users to manage their own profiles and Transport Coordinators to manage permitted operational records.		
Preconditions	**1.** The user is authenticated with an active account. **2.** The user is authorised to access the selected record.		
Postconditions	**1.** Permitted record changes are saved. **2.** Any confirmed Caregiver–Patient relationship is stored.		
Priority	High		
Frequency of Use	As required.		
Flow of Events	**1.** User selects their profile or an authorised operational record. **2.** System displays the record. **3.** User views or makes permitted changes. **4.** For Caregiver–Patient linking, system performs UC-02.1. **5.** System validates and saves the changes.		
Alternative Flows	**UC-02.AF1 - Cancel record changes:** The user may cancel the operation or exit without saving, in which case no changes are made to the record.		
Exceptions	**UC-02.EX1 - Invalid record data:** If required information is invalid or incomplete, the system rejects the changes and the record is not updated.		
Includes	Manage Patient, Caregiver, Driver and Vehicle Records; UC-02.1 Link Caregiver to Patient.		
Special Requirements	Users edit only their own personal details. Coordinators may view Patient records and edit only Status and Notes. Resource management requires Coordinator authorisation.		
Assumptions	Required record information is available.		
Notes and Issues	Related to FR-1.1, FR-1.4 and FR-1.5.		

UC-02.1: Link Caregiver to Patient

Use Case ID	UC-02.1		
Use Case Name	Link Caregiver to Patient		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	29/09/2026	**Date Last Updated**	20/10/2026
Actor	Patient, Caregiver, Transport Coordinator		
Description	Allows a Patient to authorise a Caregiver to access and perform permitted transport-related functions for them. The Caregiver generates a temporary invitation code, which the Patient enters and confirms in their own account.		
Preconditions	**1.** Patient and Caregiver accounts and profiles exist and are active. **2.** Patient acts on their own profile, or an authorised Coordinator assists.		
Postconditions	**1.** An active Caregiver–Patient relationship is stored. **2.** The Caregiver is authorised to access the linked Patient's permitted transport information and functions. **3.** The invitation code is invalidated following successful linking.		
Priority	High		
Frequency of Use	As required when Caregiver access is established or updated.		
Flow of Events	**1.** The Caregiver opens Caregiver-Patient linking in their account. **2.** The Caregiver requests an invitation code. **3.** The system generates a unique, temporary, single-use code and displays its expiry. **4.** The Caregiver shares the code with the intended Patient. **5.** The Patient opens Caregiver linking in their own account and enters the code, confirming the link.		
Alternative Flows	**UC-02.1.AC.1 - Existing relationship:** If an active relationship already exists between the selected Caregiver and Patient, the system informs the Patient and does not create a duplicate. **UC-02.1.AC.2 - Coordinator-assisted linking:** If the Patient cannot complete self-service linking, an authenticated and authorised Transport Coordinator selects the relevant active records, verifies the Patient's authorisation, and confirms the relationship. **UC-02.1.AC.3 - Unlink relationship:** Patient removes Caregiver access, or Caregiver leaves the relationship.		

Exceptions	**UC-02.1.EX.1 - Invalid invitation code:** If the code is incorrect, expired, revoked or already used, the system rejects the request without granting access. **UC-02.1.EX.2 - Ineligible account:** Inactive accounts or profiles cannot establish a link.
Includes	None.
Special Requirements	Caregiver linking is optional. Access requires a current link. Unlinking revokes access without deleting records. Code attempts are rate-limited and relationship changes are recorded.
Assumptions	Caregiver and Patient profile information is current and accurate.
Notes and Issues	Included by UC-02 Manage Profiles & Resources. Related to FR-1.4 and FR-1.5.

UC-03: Manage Transport Schedules

Use Case ID	UC-03		
Use Case Name	Manage Transport Schedules		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	20/10/2026
Actor	Patient, Caregiver, Transport Coordinator		
Description	Patients or linked Caregivers record treatment requirements. Coordinators arrange Driver transport timings and resource assignments.		
Preconditions	**1.** User is authenticated and authorised for the Patient. **2.** Patient profile and DialysisCentre reference data exist. **3.** Resource information is available when assignment is attempted.		
Postconditions	**1.** Treatment requirements are saved. **2.** When transport is arranged, outbound and return Trips are registered. **3.** Suitable resources are assigned where available; unmatched Trips remain Unfilled.		
Priority	High		
Frequency of Use	Whenever recurring transport schedules are created or updated.		
Flow of Events	**1.** Patient or linked Caregiver enters the centre, treatment days, session times and effective dates. **2.** System validates and saves the TreatmentSlot information. **3.** Coordinator initiates transport scheduling. **4.** System registers outbound and return Trips using the TreatmentSlot and standard buffer. **5.** System performs UC-03.1 Assign Driver and Vehicle. **6.** System stores transport timings and assignments, identifying unmatched Trips as Unfilled.		
Alternative Flows	**UC-03.AC.1 – Update recurring schedule:** If an existing TreatmentSlot or recurring transport requirement changes, the Transport Coordinator updates the schedule and the system recalculates the affected Trips and assignments. **UC-03.AC.2 - Coordinator-Assisted entry:** Coordinator records treatment requirements on the Patient's behalf.		
Exceptions	**UC-03.EX1 - Invalid scheduling data:** If required Patient, TreatmentSlot, Driver, Vehicle or reference data is missing or invalid, the affected Trip cannot be scheduled.		
Includes	Register Recurring Trips, UC-03.1 Assign Driver and Vehicle		

Special Requirements	Treatment submission does not confirm transport. Treatment times remain separate from pickup times. Scheduling respects the TreatmentSlot, standard 15-minute buffer and mandatory assignment constraints before Driver preferences. Completed Trip history is preserved.
Assumptions	Treatment information, Driver availability, resource information and Patient preferences are current.
Notes and Issues	Related to FR-2.1 to FR-2.7 and FR-3.3.

UC-03.1: Assign Driver and Vehicle

Use Case ID	UC-03.1		
Use Case Name	Assign Driver and Vehicle		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	29/09/2026	**Date Last Updated**	29/09/2026
Actor	Transport Coordinator		
Description	Assigns a suitable Driver and Vehicle to a Trip according to mandatory scheduling constraints, Patient Driver preference and Driver continuity preference.		
Preconditions	**1.** A Trip requiring assignment exists. **2.** Driver availability information is available. **3.** Vehicle and Patient transport requirement information is available.		
Postconditions	**1.** A suitable Driver and Vehicle are assigned to the Trip and the assignment is recorded. **2.** If no suitable assignment exists, the Trip is identified as Unfilled.		
Priority	High		
Frequency of Use	Whenever a Trip requires an initial or replacement Driver and Vehicle assignment.		
Flow of Events	**1.** System retrieves available Drivers and Vehicles for the Trip. **2.** System checks each candidate assignment against mandatory availability, time-window, mobility, equipment, capacity and assistance requirements. **3.** System checks whether the Patient has designated a preferred Driver. **4.** If the preferred Driver satisfies all mandatory constraints, the system prioritises that Driver. **5.** Otherwise, the system applies Driver continuity preference where feasible. **6.** System selects a suitable Driver and compatible Vehicle. **7.** System stores the assignment and records the assignment in the Trip history.		
Alternative Flows	**UC-03.1.AC.1 - Preferred Driver unavailable:** If the Patient's preferred Driver does not satisfy the mandatory constraints, the system continues with continuity preference and other suitable Drivers. **UC-03.1.AC.2 - Continuity Driver unavailable:** If a previously used Driver cannot satisfy the mandatory constraints, the system evaluates other suitable Drivers.		

	UC-03.1.AC.3 - No suitable assignment: If no suitable Driver and Vehicle combination exists, the system identifies the Trip as Unfilled.
Exceptions	**UC-03.1.EX.1 - Assignment data unavailable:** If required Driver, Vehicle, Patient or Trip information is missing or invalid, the system cannot complete the assignment.
Includes	Check Assignment Suitability, Check Patient Driver Preference, Apply Driver Continuity Preference, Identify Unfilled Trips, Record Trip History.
Special Requirements	Mandatory scheduling constraints must be satisfied before Patient Driver preference or Driver continuity preference is applied. Patient-designated preferred Driver takes precedence over continuity preference where both are feasible.
Assumptions	Driver availability, Vehicle information, Patient transport requirements and Driver preferences are current.
Notes and Issues	Included by UC-03 Manage Transport Schedules. Related to FR-2.2 to FR-2.7 and FR-3.3.

UC-04: Monitor Transport Operations

Use Case ID	UC-04		
Use Case Name	Monitor Transport Operations		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Transport Coordinator		
Description	Allows the Transport Coordinator to monitor scheduled Trips and their current operational status through the Coordinator Dashboard.		
Preconditions	**1.** The Transport Coordinator is authenticated. **2.** Trip records exist.		
Postconditions	**1.** Current Trip statuses are displayed. **2.** Unresolved At-Risk or Unfilled Trips remain visible.		
Priority	High		
Frequency of Use	Frequently during operating hours.		
Flow of Events	**1.** Coordinator opens the dashboard. **2.** System retrieves current Trips and statuses. **3.** System categorises Trips as On-Track, At-Risk, Unfilled or Resolved. **4.** Coordinator reviews the required category or Trip.		
Alternative Flows	**UC-04.AF1 - Filter or inspect trips:** The Transport Coordinator may filter the dashboard by Trip status or select an individual Trip to view additional information. **UC-04.AF2 - View audit log:** The Transport Coordinator opens the Audit Log from the Dashboard and reviews the recorded Trip history, filtering by entity type, actor or action.		
Exceptions	**UC-04.EX1 - Trip data unavailable:** If current Trip information cannot be retrieved due to a temporary system or service interruption, the latest Trip statuses cannot be displayed.		
Includes	None.		
Special Requirements	Trips classified as At-Risk or Unfilled must remain visible on the dashboard until resolved.		
Assumptions	Trip status information is kept current by the system.		
Notes and Issues	Related to FR-6.1 and FR-6.5.		

UC-05: Resolve At-Risk Trip

Use Case ID	UC-05		
Use Case Name	Resolve At-Risk Trip		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Transport Coordinator		
Description	Allows the Transport Coordinator to review ranked recovery options and approve a suitable response for an At-Risk Trip.		
Preconditions	1. A Trip has been classified as At-Risk. 2. Current information required for recovery assessment is available.		
Postconditions	1. An approved recovery action is applied and the Trip assignment is modified. 2. If no feasible recovery exists, the Trip is escalated as Unfilled-Critical.		
Priority	High		
Frequency of Use	Whenever a Trip becomes At-Risk.		
Flow of Events	1. System detects an At-Risk Trip. 2. System generates and ranks feasible recovery options. 3. Transport Coordinator reviews the ranked recovery options. 4. Transport Coordinator selects a proposed recovery option. 5. The system performs UC-05.1 Approve Recovery Action. 6. Following approval, the system modifies the Trip assignment and triggers notifications to affected users. 7. System updates the Trip status accordingly.		
Alternative Flows	None.		
Exceptions	**UC-05.EX1 - No feasible recovery option:** If no recovery option is available, the system flags the Trip as Unfilled-Critical and escalates it for manual intervention.		
Includes	Generate and Rank Recovery Options, UC-05.1 Approve Recovery Action.		
Special Requirements	Recovery options must be generated within 10 seconds after disruption detection.		
Assumptions	Current Driver, Vehicle, route and traffic information is available for recovery assessment.		
Notes and Issues	Related to FR-5.1 to FR-5.5 and NFR-2.2.		

UC-05.1: Approve Recovery Action

Use Case ID	UC-05.1		
Use Case Name	Approve Recovery Action		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	29/09/2026	**Date Last Updated**	29/09/2026
Actor	Transport Coordinator		
Description	Allows the Transport Coordinator to approve a feasible recovery option before an At-Risk Trip assignment is modified.		
Preconditions	**1.** The Trip is classified as At-Risk. **2.** The system has generated one or more feasible recovery options.		
Postconditions	**1.** The selected recovery action is recorded as approved. **2.** The corresponding Trip modification is applied. **3.** The approved recovery action and resulting reassignment are recorded in the Trip history.		
Priority	High		
Frequency of Use	Whenever a feasible recovery option is selected for an At-Risk Trip.		
Flow of Events	**1.** System displays the ranked feasible recovery options for the At-Risk Trip. **2.** Transport Coordinator reviews the proposed recovery options. **3.** Transport Coordinator selects a recovery option. **4.** System displays the proposed changes to the Trip. **5.** Transport Coordinator approves the recovery action. **6.** System records the approval. **7.** System modifies the Trip assignment according to the approved recovery action. **8.** System records the recovery action and reassignment in the Trip history.		
Alternative Flows	**UC-05.1.AC.1 - Select another recovery option:** The Transport Coordinator does not approve the selected option and returns to the available recovery options to select another feasible option.		
Exceptions	**UC-05.1.EX.1 - Recovery option no longer feasible:** If the selected recovery option becomes infeasible before approval, the system rejects the approval and requires the available recovery options to be reassessed.		
Includes	Modify Trip Assignment.		
Special Requirements	A Trip assignment must not be modified by a proposed recovery option until the Transport Coordinator approves that option.		

Assumptions	Recovery options are generated using sufficiently current Driver, Vehicle, route and traffic information.
Notes and Issues	Included by UC-05 Resolve At-Risk Trip. Related to FR-5.3, FR-5.5 and FR-6.4.

UC-06: Manage Driver Availability

Use Case ID	UC-06		
Use Case Name	Manage Driver Availability		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Driver		
Description	Allows a Driver to declare, view and update schedule availability for transport assignments.		
Preconditions	**1.** The Driver is authenticated. **2.** The Driver has a verified Driver account.		
Postconditions	**1.** The Driver's Availability **2.** Window information is stored or updated.		
Priority	High		
Frequency of Use	As required by the Driver.		
Flow of Events	**1.** Driver opens availability management. **2.** System displays existing availability. **3.** Driver adds or updates an availability period. **4.** System validates and saves the change.		
Alternative Flows	**UC-06.AF1 - View availability only:** The Driver may view their existing availability and exit without making any changes.		
Exceptions	**UC-06.EX1 - Invalid availability:** If the entered availability is invalid or conflicts with existing availability information, the system rejects the change.		
Includes	None.		
Special Requirements	Availability information used for scheduling must be kept current.		
Assumptions	The Driver provides accurate availability information.		
Notes and Issues	Related to FR-3.1.		

UC-07: Report Driver Unavailability

Use Case ID	UC-07		
Use Case Name	Report Driver Unavailability		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Driver		
Description	Allows an assigned Driver to report that they can no longer perform an assigned Trip.		
Preconditions	**1.** The Driver is authenticated. **2.** The Driver has an assigned active or upcoming Trip.		
Postconditions	**1.** The Driver's unavailability is recorded. **2.** The affected Trip is automatically classified as At-Risk.		
Priority	High		
Frequency of Use	Occasionally, when unexpected unavailability occurs.		
Flow of Events	**1.** Driver selects the affected assigned Trip. **2.** Driver reports sudden unavailability. **3.** Driver confirms the report. **4.** System records the unavailability report and corresponding disruption in the Trip history. **5.** System flags the Trip as At-Risk.		
Alternative Flows	**UC-07.AF1 - Cancel unavailability report:** The Driver may cancel the report before it is submitted, in which case no changes are made to the Trip.		
Exceptions	**UC-07.EX1 - Trip no longer eligible:** If the selected Trip has already been completed or cancelled, the Driver cannot report sudden unavailability for that Trip.		
Includes	None.		
Special Requirements	The affected At-Risk Trip must remain visible to the Transport Coordinator until resolved.		
Assumptions	The Driver reports unexpected unavailability as soon as it becomes known.		
Notes and Issues	Related to FR-3.2, FR-5.1, FR-6.4 and FR-6.5.		

UC-08: Assess Route and ETA

Use Case ID	UC-08		
Use Case Name	Assess Route and ETA		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	OneMap API, LTA DataMall		
Description	Calculates a Trip's route and estimated arrival time using baseline routing information together with near-real-time traffic conditions.		
Preconditions	**1.** The Trip origin and destination are known. **2.** The required external services are available.		
Postconditions	**1.** A RouteAssessment containing the route, distance, traffic conditions and recalculated ETA is available.		
Priority	High		
Frequency of Use	On demand and when route conditions require reassessment.		
Flow of Events	**1.** System retrieves a baseline route, distance and travel-time estimate from OneMap. **2.** System retrieves traffic incidents and road speed bands from LTA DataMall. **3.** System maps relevant traffic conditions onto the route. **4.** System recalculates the ETA.		
Alternative Flows	**UC-08.AF1 - No relevant traffic disruption:** If no relevant traffic disruption affects the Trip route, the baseline route and travel-time estimate remain applicable.		
Exceptions	**UC-08.EX1 - External service failure:** If OneMap or LTA DataMall is unavailable, times out or returns unusable data, the RouteAssessment cannot be fully completed using current external data.		
Includes	Retrieve Baseline Route, Retrieve Traffic Conditions, Map Traffic to Route, Recalculate ETA.		
Special Requirements	Route and travel-time information must be returned within 5 seconds under normal operating conditions. External integrations must remain separate from core scheduling logic.		
Assumptions	External mapping and traffic services provide sufficiently current and accurate information.		
Notes and Issues	Related to FR-4.1 to FR-4.4, NFR-2.1 and NFR-7.1.		

UC-09: Track Active Trip

Use Case ID	UC-09		
Use Case Name	Track Active Trip		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Transport Coordinator, Patient, Caregiver		
Description	Allows authorised users to view the assigned Driver's live location and Trip progress while an assigned Trip is active.		
Preconditions	**1.** The Trip is active. **2.** A Driver is assigned to the Trip. **3.** The viewing user is authorised to access the Trip. **4.** If the viewing user is a Caregiver, the Caregiver is explicitly linked to the relevant Patient.		
Postconditions	**1.** Driver location and Trip progress are displayed while the Trip remains active. **2.** Driver location sharing stops when the Trip is completed or cancelled.		
Priority	High		
Frequency of Use	During active Trips.		
Flow of Events	**1.** Authorised user opens an active Trip. **2.** System obtains the assigned Driver's current location. **3.** System displays the Driver on a OneMap-based map together with Trip progress. **4.** Location and progress update while the Trip remains active. **5.** System stops sharing the Driver’s location when the Trip ends.		
Alternative Flows	**UC-09.AF1 - Exit tracking view:** The authorised user may exit the tracking view while the Trip remains active, the Trip continues normally.		
Exceptions	**UC-09.EX1 - Live location unavailable:** If the Driver's current location cannot be retrieved, the system cannot display the Driver's current live location.		
Includes	Display Driver Location and Trip Progress.		
Special Requirements	Driver live location may only be shared with authorised users during an active assigned Trip. A Caregiver may only access tracking information for linked Patients.		
Assumptions	The Driver's device is capable of providing location information during the active Trip, and any Caregiver accessing the Trip has a valid Caregiver-Patient link.		

Notes and Issues	Related to FR-1.5, FR-4.5 to FR-4.7 and NFR-6.2.

UC-10: Notify Affected Users

Use Case ID	UC-10		
Use Case Name	Notify Affected Users		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Driver, Patient, Caregiver		
Description	Notifies affected users when a Transport Coordinator approves a modification to a Trip.		
Preconditions	**1.** A Trip modification has been approved by a Transport Coordinator. **2.** The affected users can be identified.		
Postconditions	**1.** A notification containing the relevant updated transport information is issued to each affected user.		
Priority	High		
Frequency of Use	Whenever an approved Trip modification affects one or more users.		
Flow of Events	**1.** Coordinator approves a Trip modification. **2.** System identifies the affected Driver, Patient and/or authorised Caregiver linked to the Patient. **3.** System generates the required notification. **4.** System sends the notification to the relevant users.		
Alternative Flows	**UC-10.AF1 - Notify selected users only:** If only some users are affected by the Trip modification, the system sends notifications only to those affected users.		
Exceptions	**UC-10.EX1 - Notification delivery failure:** If delivery to an affected user fails, the notification cannot be successfully delivered. **UC-10.EX2 - Missing recipient information:** If required recipient contact information is unavailable, the system cannot send the notification to that user.		
Includes	Send Notifications to Driver, Patient and Caregiver.		
Special Requirements	Notifications must reveal only information appropriate to the recipient's role. Caregivers may only receive Patient-specific Trip information for Patients to whom they are explicitly linked.		
Assumptions	Valid notification or contact details are available for affected users, and any Caregiver recipient is linked to the relevant Patient.		
Notes and Issues	Related to FR-1.5, FR-6.2 and the role/privacy constraints in NFR-5.2 and NFR-6.1.		

UC-11: Manage Upcoming Pickup

Use Case ID	UC-11		
Use Case Name	Manage Upcoming Pickup		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	29/09/2026
Actor	Patient, Caregiver		
Description	Allows a Patient or authorised Caregiver to view an upcoming pickup and, where applicable, confirm or cancel the scheduled Trip.		
Preconditions	**1.** The Patient or Caregiver is authenticated. **2.** The user is authorised to access the relevant Patient's transport information. **3.** If the user is a Caregiver, the Caregiver is explicitly linked to the relevant Patient.		
Postconditions	**1.** The upcoming pickup details are displayed. **2.** If the Trip is confirmed, the confirmation is recorded. **3.** If the Trip is cancelled, its status is changed to Cancelled. **4.** If the Trip is cancelled, its assigned Driver and Vehicle capacity is released.		
Priority	High		
Frequency of Use	As required, typically before a scheduled Trip.		
Flow of Events	**1.** Patient or Caregiver opens upcoming pickup information. **2.** System retrieves the Patient's next scheduled Trip. **3.** System displays the pickup time, assigned Driver, Trip status and relevant delay information. **4.** System displays the actions available for the upcoming Trip.		
Alternative Flows	**UC-11.AF1 - No upcoming Trip:** If no upcoming Trip exists, the system informs the user that no pickup is currently scheduled. **UC-11.AF2 - Confirm upcoming Trip:** The Patient or Caregiver confirms the Trip and the system records the confirmation. **UC-11.AF3 - Cancel upcoming Trip:** The Patient or Caregiver chooses to cancel the upcoming Trip. The system proceeds with UC-11.1 Cancel Upcoming Trip.		
Exceptions	**UC-11.EX1 - Pickup data unavailable:** If upcoming Trip information cannot be retrieved due to a temporary system failure, the pickup details cannot be displayed.		

	UC-11.EX.2 - Unauthorised Caregiver: If the Caregiver is not linked to the relevant Patient, access to the Patient's Trip information is denied.
Includes	View Pickup Details.
Special Requirements	The next scheduled pickup must be reachable within three user interactions after login. Trip information must meet WCAG 2.1 AA contrast and readability requirements. Caregiver access is limited to explicitly linked Patients.
Assumptions	Caregiver-Patient links are current and valid.
Notes and Issues	Extended by UC-11.1 Cancel Upcoming Trip. Related to FR-1.5, FR-5.6 to FR-5.9, FR-6.3, FR-6.6, NFR-1.1 and NFR-1.2.

UC-11.1: Cancel Upcoming Trip

Use Case ID	UC-11.1		
Use Case Name	Cancel Upcoming Trip		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	29/09/2026	**Date Last Updated**	29/09/2026
Actor	Patient, Caregiver		
Description	Allows a Patient or authorised Caregiver to cancel an upcoming Trip before its scheduled pickup, releasing the assigned Driver and Vehicle capacity for potential reassignment.		
Preconditions	**1.** The Patient or Caregiver is authenticated. **2.** The user is authorised to access the relevant Patient's Trip. **3.** If the user is a Caregiver, the Caregiver is explicitly linked to the relevant Patient. **4.** The Trip is upcoming and has not started.		
Postconditions	**1.** The Trip is marked as Cancelled. **2.** The cancellation is recorded in the Trip history. **3.** The assigned Driver and Vehicle capacity is released. **4.** Released capacity may be considered for reassignment to an Unfilled Trip.		
Priority	High		
Frequency of Use	Occasionally, when an upcoming Trip is no longer required.		
Flow of Events	**1.** Patient or Caregiver opens the upcoming Trip. **2.** User selects Cancel Upcoming Trip. **3.** System displays the Trip details and requests confirmation. **4.** User confirms the cancellation. **5.** System validates that the Trip remains eligible for cancellation. **6.** System marks the Trip as Cancelled. **7.** System records the cancellation in the Trip history. **8.** System releases the assigned Driver and Vehicle capacity. **9.** System identifies whether the released capacity may satisfy an Unfilled Trip.		
Alternative Flows	**UC-11.1.AC.1 - Cancel cancellation request:** The Patient or Caregiver chooses not to confirm the cancellation. The Trip remains unchanged.		
Exceptions	**UC-11.1.EX.1 - Trip no longer eligible for cancellation:** If the Trip is already active, completed or cancelled, the system rejects the cancellation request.		

	UC-11.1.EX.2 - Unauthorised Caregiver: If the Caregiver is not linked to the relevant Patient, the system denies access to the Trip and does not process the cancellation.
Includes	None.
Special Requirements	Cancellation is only permitted before the scheduled pickup. Caregivers may only cancel Trips belonging to Patients to whom they are explicitly linked.
Assumptions	The Trip status and Caregiver-Patient authorisation information are current.
Notes and Issues	Extends UC-11 Manage Upcoming Pickup. Related to FR-1.5, FR-5.6 to FR-5.9 and FR-6.4. If a linked Google Calendar reminder exists, its removal is handled through UC-13 Manage Calendar Reminders.

UC-12: Manage Driver Feedback and Preference

Use Case ID	UC-12		
Use Case Name	Manage Driver Feedback and Preference		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Patient		
Description	Allows a Patient to rate a Driver after a completed Trip and manage their preferred Driver for future transport assignments.		
Preconditions	1. The Patient is authenticated. 2. Driver rating and Trip assignment records are available.		
Postconditions	1. A submitted Driver rating is stored, where applicable. 2. The Patient's Driver preference is stored, updated or removed, where applicable.		
Priority	Medium		
Frequency of Use	After completed Trips or whenever the Patient wishes to change their preferred Driver.		
Flow of Events	1. Patient opens Driver feedback and preference management. 2. System displays eligible Drivers and relevant completed Trips. 3. Patient selects whether to rate a Driver or manage their preferred Driver. 4. Patient submits the rating or Driver preference. 5. System validates and stores the submitted information.		
Alternative Flows	**UC-12.AF1 - Update preferred Driver:** The Patient replaces their existing preferred Driver with another eligible Driver. **UC-12.AF2 - Remove preferred Driver:** The Patient removes their existing Driver preference without selecting a replacement.		
Exceptions	**UC-12.EX1 - Ineligible rating:** If the selected Trip was not completed by the selected Driver for that Patient, the system rejects the rating. **UC-12.EX2 - Invalid preference:** If the selected Driver is not eligible to be designated as a preferred Driver, the system does not save the preference.		
Includes	Rate Driver, Set Preferred Driver.		
Special Requirements	Driver preferences must remain secondary to mandatory scheduling, mobility, capacity and assistance requirements.		

Assumptions	Completed Trip and Driver assignment records are available for rating validation.
Notes and Issues	Related to FR-2.6, FR-2.7, FR-7.1 and FR-7.2.

UC-13: Manage Calendar Reminders

Use Case ID	UC-13		
Use Case Name	Manage Calendar Reminders		
Created By	Joshua Chin Wei Jie	**Last Updated By**	Joshua Chin Wei Jie
Date Created	10/09/2026	**Date Last Updated**	13/09/2026
Actor	Patient, Caregiver		
Description	Allows Patients and authorised Caregivers to create and maintain Google Calendar reminders for scheduled transport pickups.		
Preconditions	**1.** The user is authenticated. **2.** The user has access to an upcoming scheduled Trip. **3.** The user has authorised access to their Google Calendar. **4.** If the user is a Caregiver, the Caregiver is explicitly linked to the relevant Patient.		
Postconditions	**1.** A corresponding Google Calendar reminder is created, updated or removed to reflect the current Trip schedule. **2.** The reminder synchronisation status is recorded.		
Priority	Medium		
Frequency of Use	When a user enables a reminder or when a linked Trip is subsequently modified or cancelled.		
Flow of Events	**1.** Patient or Caregiver selects an upcoming Trip. **2.** User chooses to create a calendar reminder. **3.** System sends the required Trip information to Google Calendar. **4.** Google Calendar creates the linked event. **5.** System stores the external event reference and synchronisation status. **6.** If the Trip is later modified or cancelled, the system updates or removes the linked calendar event accordingly.		
Alternative Flows	**UC-13.AF1 - Update linked reminder:** If the corresponding Trip pickup time changes, the system updates the existing calendar event. **UC-13.AF2 - Remove linked reminder:** If the corresponding Trip is cancelled, the system removes or cancels the linked calendar event.		
Exceptions	**UC-13.EX1 - Calendar service unavailable:** If Google Calendar is unavailable, the reminder cannot be synchronised, but core Trip scheduling and monitoring remain unaffected. **UC-13.EX2 - Calendar authorisation unavailable:** If the user's Google Calendar authorisation is missing, expired or revoked, the system cannot synchronise the reminder.		

Includes	Sync Pickup Reminder.
Special Requirements	The system must obtain explicit user authorisation before accessing Google Calendar, must only share Trip information required for reminder synchronisation, and must restrict Caregiver reminders to linked Patients.
Assumptions	The user has a valid Google account, Google Calendar is reachable when synchronisation occurs, and any Caregiver accessing the Trip has a valid Caregiver-Patient link.
Notes and Issues	Related to FR-1.5, FR-7.3 to FR-7.5, NFR-3.2, NFR-5.3, NFR-6.3 and NFR-7.1.

Use Case Template

Use Case ID:			
Use Case Name:			
Created By:		**Last Updated By:**	
Date Created:		**Date Last Updated:**	

Actor:	
Description:	
Preconditions:	
Postconditions:	
Priority:	
Frequency of Use:	
Flow of Events:	
Alternative Flows:	
Exceptions:	
Includes:	
Special Requirements:	
Assumptions:	
Notes and Issues:	